

Chapter 1 Review

Renardy & Rogers: *An Introduction to PDEs*

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Purpose of the Review

- Fix notation and language for PDEs
- Review ODEs, linear algebra, and calculus tools
- Motivate structural differences between PDE types

Partial Differential Equations

- Unknown function $u = u(x_1, \dots, x_n)$
- Involves partial derivatives of u
- General form:

$$F(x, u, Du, D^2u, \dots) = 0$$

- Order = highest derivative appearing

Linear vs Nonlinear PDEs

Linear PDE:

$$Lu = f, \quad L = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$$

- Superposition principle
- Homogeneous vs inhomogeneous

Nonlinear PDE:

- Nonlinear dependence on u or derivatives
- Qualitatively new phenomena

Model PDEs

- Transport: $u_t + cu_x = 0$
- Heat: $u_t - \kappa \Delta u = 0$
- Wave: $u_{tt} - c^2 \Delta u = 0$
- Laplace: $\Delta u = 0$

Classification of Second-Order PDEs

For

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} + \dots = 0$$

Discriminant: $B^2 - AC$

- Elliptic: < 0 (steady state)
- Parabolic: $= 0$ (diffusion)
- Hyperbolic: > 0 (propagation)

Initial and Boundary Conditions

- PDEs require auxiliary data
- Initial conditions (time-dependent problems)
- Boundary conditions:
 - Dirichlet
 - Neumann
 - Robin / mixed

Well-Posedness (Hadamard)

A problem is well-posed if:

- ① A solution exists
- ② The solution is unique
- ③ Continuous dependence on data

Well-posed is natural question in PDE.

- Ordinary differential equations as 1D PDEs
- Initial value problems:

$$u'(t) = f(t, u), \quad u(t_0) = u_0$$

- Existence and uniqueness via Lipschitz conditions

Linear Constant Coefficient ODEs

$$u'(t) = Au(t), \quad A \in \mathbb{R}^{n \times n}$$

- Solution: $u(t) = e^{tA}u(0)$
- Behavior governed by eigenvalues of A
- Growth, decay, oscillation

Eigenvalues and Stability

- $\operatorname{Re}(\lambda) < 0 \Rightarrow$ decay
- $\operatorname{Re}(\lambda) > 0 \Rightarrow$ growth
- Imaginary eigenvalues $\Rightarrow$ oscillation

PDE behavior often mirrors infinite-dimensional versions of this picture.

Linear Algebra Review

- Vector spaces and linear operators
- Eigenvalues and eigenvectors
- Diagonalization and spectral decomposition

Inner Products and Norms

- Inner products induce norms
- $L^2(\Omega)$ as a natural energy space
- Orthogonality and expansions

$$\|u\|^2 = \langle u, u \rangle$$

Multivariable Calculus Tools

- Gradient, divergence, Laplacian
- Chain rule and change of variables
- Scaling and nondimensionalization

Integration by Parts and Divergence Theorem

Key identity:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = - \int_{\Omega} u \Delta v \, dx + \int_{\partial\Omega} u \partial_n v \, dS$$

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- Foundation for energy methods
- Gateway to weak formulations

Key Takeaways

- PDEs generalize ODEs in space
- Classification predicts behavior
- Linear algebra and ODE intuition persist
- Analysis and geometry are inseparable

Looking Ahead

- Explicit solution techniques
- Fourier methods
- Weak solutions and functional analysis